

Semantic Search for Nepali Municipal Legal Documents

Hybrid PDF Extraction, Chunking, and Retrieval Evaluation

Student Research Presentation

Department of Computer Science

Six-Minute Beamer Presentation

Outline

- 1 Background
- 2 Aim
- 3 Literature
- 4 Methods
- 5 Evaluation
- 6 Results
- 7 Paper Outline
- 8 Conclusion

Background and Motivation

- Kathmandu Valley laws are published across municipal websites as PDF files.
- The collection includes national laws, municipal bylaws, procedures, and financial acts.
- Many users need simple procedural answers, but keyword search across PDFs is weak.
- A retrieval system must first extract reliable Nepali text, split it into useful chunks, and evaluate whether search returns the right legal section.

Aim and Objectives

Aim

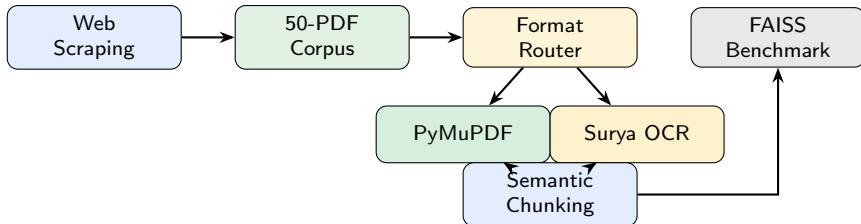
Build and evaluate a small retrieval-ready corpus for Nepali municipal legal documents.

- ① Scrape and curate a focused corpus of legal PDFs.
- ② Route text-based PDFs and scanned PDFs through suitable extractors.
- ③ Convert extracted text into semantic chunks.
- ④ Benchmark dense retrieval using query-level ground truth.
- ⑤ Identify practical errors that affect retrieval quality.

Literature Review

- Legal information retrieval requires high precision because answers depend on exact sections and conditions.
- Retrieval-augmented generation depends on document preprocessing: extraction quality and chunk boundaries directly affect final answers.
- Multilingual embedding models are usually stronger for Nepali and code-switched queries than English-only baselines.
- Prior OCR work shows that scanned Devanagari documents need separate quality tracking because recognition errors become retrieval errors.

Method: End-to-End Pipeline



The pipeline produces raw text, JSONL chunks, extraction summaries, and retrieval metrics.

Curated corpus

- 50 legal PDFs
- 830 total pages
- Kathmandu, Lalitpur, Bhaktapur, and federal sources
- 45 text-based, 5 scanned

Current outputs

- 50 chunk files
- 446 total chunks
- 11 evaluation queries
- MiniLM baseline CSV generated

Evaluation Method

- Each query has a target legal chunk as ground truth.
- Retrieval is evaluated with **Recall@5** and **MRR**.
- FAISS performs cosine-similarity search over embedding vectors.
- Baseline model: `sentence-transformers/all-MiniLM-L6-v2`.
- Multilingual model planned: `BAAI/bge-m3`.

Why this matters

If the correct section is not retrieved, a later RAG answer cannot cite the right legal basis.

Results and Findings

Run	Recall@5	MRR
MiniLM baseline, 11 queries	0.000	0.000
BGE-M3 multilingual run	Pending	Pending

- The baseline result is not a final failure of the idea; it reveals evaluation and language mismatch.
- Ground-truth chunk IDs required mapping before fair comparison.
- Some scanned documents produce sparse chunks, showing OCR quality is a retrieval risk.

Paper Outline

- ➊ Introduction: problem, motivation, and objectives.
- ➋ Related Work: legal retrieval, RAG preprocessing, multilingual embeddings, OCR.
- ➌ Methodology: scraping, curation, extraction routing, chunking, and benchmarking.
- ➍ Corpus: sources, document categories, format types, and limitations.
- ➎ Evaluation: query bank, metrics, and ground-truth alignment.
- ➏ Results and Discussion: baseline behavior, errors, and improvements.
- ➐ Conclusion: contributions and future work.

Conclusion

- The project builds a working retrieval-preparation pipeline for Nepali municipal legal PDFs.
- The strongest contribution is the complete path from scraped PDFs to chunked corpus and benchmarkable queries.
- The main lesson is that extraction quality, chunk IDs, and multilingual model choice are as important as the search algorithm.
- Next steps: complete OCR validation, run BGE-M3 fully, expand human-labeled queries, and test hybrid BM25 plus dense retrieval.

Questions?

Repository: `scrapktm`
Slides built with Beamer